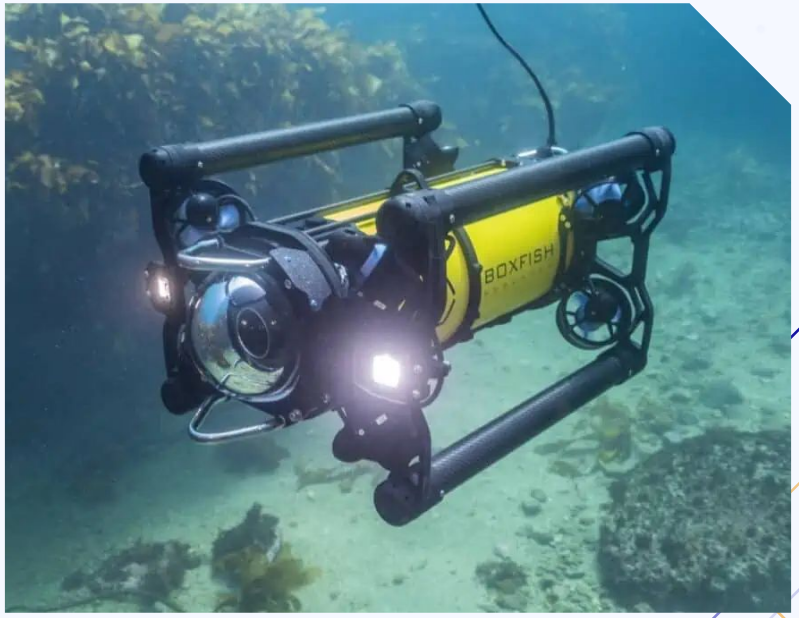
Domain Specialized BLIP-2 for Underwater Image Captioning

A Lightweight Approach for Marine Robotics

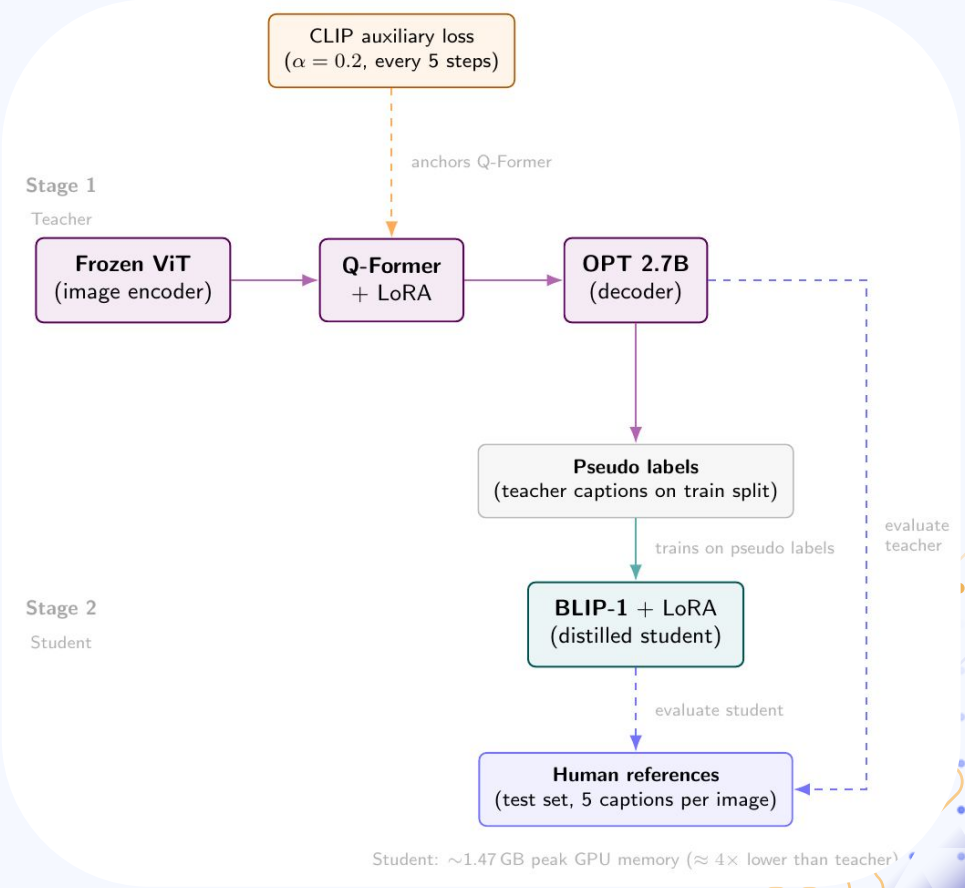
Why underwater captioning?

- **Use cases:** AUV/ROV footage, ecology, inspection. Need searchable text, not only pixels.
- **Constraints:** Often no cloud; inference must run on device with tight memory and latency.
- **Domain gap:** Color cast, turbidity, low contrast; marine subjects are rare in generic pretraining.
- **Observation:** Large encoders still see structure and texture; the harder part is domain vocabulary and precise descriptions (species, behavior).



BLIP-2 Explorer two-stage goal

Teacher: Adapt BLIP-2 OPT-2.7B to underwater captions using LoRA on the Q-Former and decoder (frozen ViT).
Student: Distill into a smaller BLIP-1 using pseudo labels from the teacher for embedded deployment.



Dataset: UICD

Scale & Split

- **3,176** JPEG images at 224×224
- **5** human-written captions each
- **80/10/10** split (seed 42) 2,540 train / 317 val / 319 test

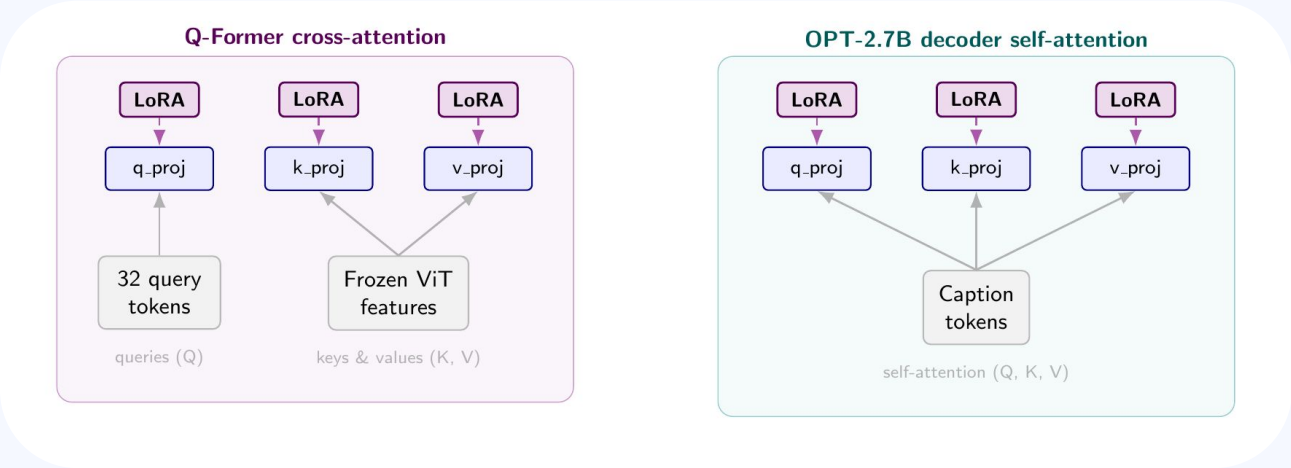
Visual Challenges

- Blue-green color cast (depth)
- Motion blur & turbidity haze
- Rare species, niche vocabulary

(a)	**GT1**: Coral reefs in seawater come in a variety of colors, such as dark-brown, light-brown, and white. **GT2**: An underwater coral reef full of life, with all kinds of corals growing freely. **GT3**: Coral reefs in the water grow corals of various shapes and colors. **GT4**: An undersea coral reef with corals of various colors and shapes. **GT5**: There are many corals of various shapes and colors growing on the underwater reefs.	(b)	**GT1**: Several small fish swim and play around the coral reef on the seafloor. **GT2**: There are some fish on the seabed, among which a blue and green fish is particularly eye-catching. **GT3**: A colorful small fish swims towards the reef at the seafloor. **GT4**: There are some corals growing on the seabed, and some small fish swim near them. **GT5**: Several small fish swim near the rocks on the seafloor.
(c)	**GT1**: A pipe is laid on the seabed and covered with fine sand. **GT2**: There are many seagrasses growing near the pipelines laid on the seabed. **GT3**: A long pipe is laid on the seabed. **GT4**: A pipe is laid on the seabed and covered with fine sand. **GT5**: There are many green seagrasses growing on the seabed, and a pipeline is also laid.	(d)	**GT1**: The clear water has many rocks of different shapes . **GT2**: There are rocks of different shapes in the seawater. **GT3**: There are many rocks of different sizes on the seabed. **GT4**: There are many rocks of different sizes and shapes on the seabed. **GT5**: An underwater scene with rocks of various sizes and shapes.
(e)	**GT1**: Corals with different shapes and colors grow on the reef under the sea . **GT2**: Colorful corals grow on the reef at the seafloor. **GT3**: Many corals grow on the reef at the seafloor. **GT4**: Corals with different shapes and colors grow on the seabed . **GT5:** Colorful corals grow on the seafloor.	(f)	**GT1**: Several small fish swim near the coral. **GT2**: Corals of various colors grow on the reefs at the seafloor. **GT3**: A clear underwater landscape with a variety of brightly colored corals. **GT4**: Corals on the seabed come in various shapes and sizes, some fan-shaped and some disc-shape. **GT5**: There are various colorful corals on the seabed, including brown-red, light brown, and green.

LoRA Adaptation

- Low-rank update W + BA on q/k/v in Q-Former cross-attention and OPT self-attention; rank r = 16, α = 16, dropout 0.1.
- Frozen: ViT and most of the LM; only LoRA paths trained.
- At inference, merge A, B into weights
  ⇒ no extra latency vs. dense layers.



Post-processing & Distillation

Period-suffix fix

- Raw model trails off: "*coral . . . . . .*"
- CIDEr collapses 1.159 → 0.017; CLIP unaffected
- Single regex fix: no retraining, applied to all models

Knowledge distillation

- Student (LoRA BLIP-1) trained on **teacher pseudo labels**
- Both evaluated on **human references** for fair comparison
- Memory: 5.5 GB → 1.47 GB (**~4× reduction**)
- CIDEr: 1.32 → 0.70. Gap partly from pseudo-label bias

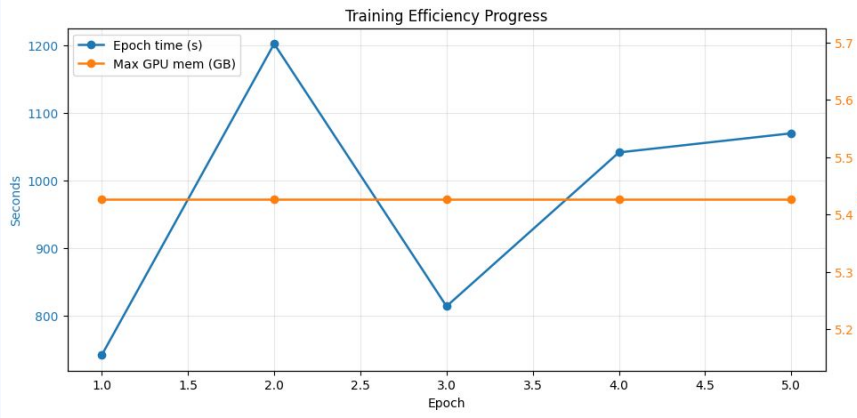
Qualitative Example

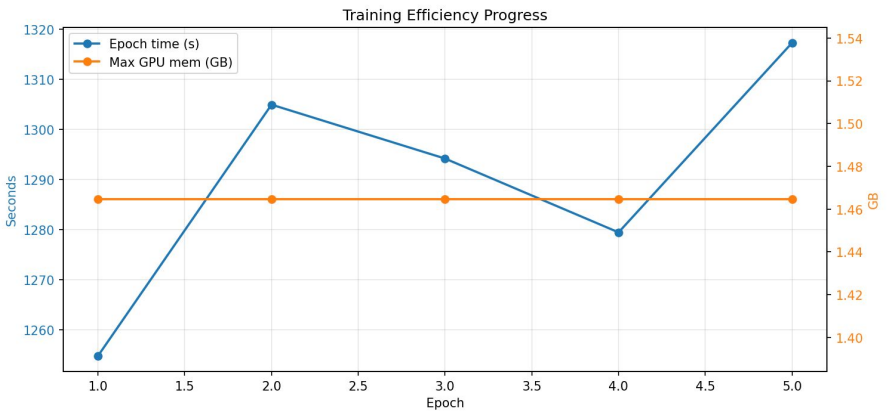
Model prediction (Distilled Model)
"A sea turtle paddles its limbs and swims above the seabed."

Ground-truth references (5 human captions):

- "A sea turtle swims in clear waters."
- "Many seagrass grows on the seabed, and a turtle swims above it."
- "A big turtle swims to the left with open arms and legs."
- "A big turtle swims above the seabed reef where seagrass grows."
- "A big turtle swims above the seabed where seagrass grows."

Training Efficiency



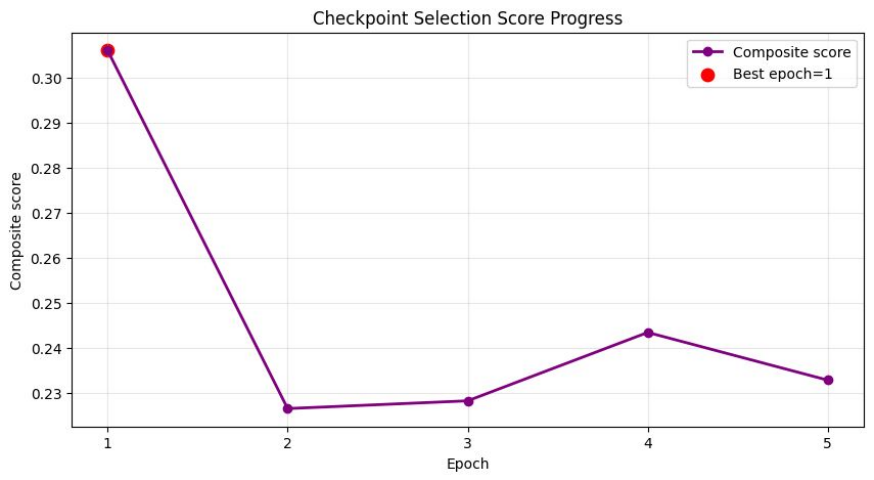


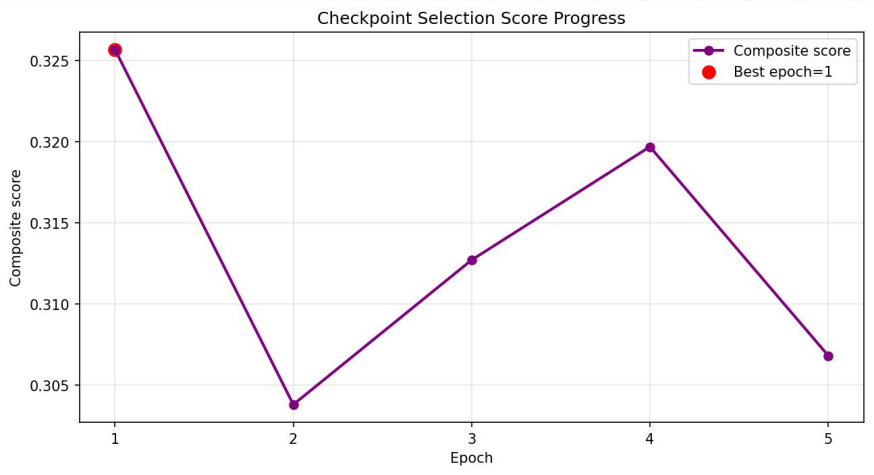
Teacher: LoRA BLIP-2 OPT-2.7B

Student: Distilled BLIP-1

Teacher: ~5.5 GB peak; Student: ~1.47 GB (~4× reduction).

Composite Score over Epochs





Teacher: LoRA BLIP-2 OPT-2.7B

Student: Distilled BLIP-1

Best composite checkpoint at epoch 1 for both teacher & student in our runs.

Key takeaways

General Result:

- **LoRA closes the vocabulary gap**, not the vision gap: CIDEr +179%, CLIP +0.6%
- **Composite checkpoint selection** prevents single-metric gaming; best epoch is 1 for both models
- **Period-suffix fix** is load-bearing: CIDEr would be 0.017 without it (×77 difference)

Student Result:

- CIDEr 1.32 → 0.70 is **not purely capacity.** Pseudo-label bias also contributes
- ~4× memory reduction (5.5 GB → 1.47 GB) with **competitive** captioning quality
- Pattern generalises: *adapt a frozen encoder,* then compress applies beyond underwater

Conclusion

BLIP-2 Explorer in a nutshell:
Adapt **BLIP-2** to underwater imagery with **LoRA** and a **CLIP** auxiliary loss, then distill into a compact **BLIP-1 student** for embedded deployment; without rewriting the architecture.

What we showed:

- Domain gap is a vocabulary problem, not a perception problem
- Multi-metric composite score prevents checkpoint overfitting
- ~4× memory reduction with a distilled student at competitive quality

Number trail to remember

CIDEr: **0.47** (base)

↓

1.32 (LoRA teacher)

↓

0.70 (distilled student)

Thanks!

Do you have any questions?